A Novel Spatial–Spectral Pyramid Network for Hyperspectral Image Classification

Junbo Zhou, Shan Zeng, *Member, IEEE*, Guoqiang Gao, Yulong Chen, and Yuanyan Tang, *Life Fellow, IEEE*

***Abstract*— As the research on deep learning methods gradually progresses, more and more classification models are applied in the classification of hyperspectral image (HSI). High-dimensional and low-resolution characteristics of HSI, however, make it difficult for conventional models to process its data effectively. In this article, a novel HSI classification model, namely, spatial–spectral pyramid network (SSPN), is designed by combining a 3-D convolutional neural network (3D CNN) with feature pyramid structure. SSPN taking advantage of 3-D convolution coupled with multiscale convolutional extraction is used to obtain a large set of diverse spatial–spectral features. Multiscale interfusion is also applied in SSPN to enrich the features contained in a single feature map and to improve the sensitivity on HSI spatial–spectral information, allowing it to better learn spatial–spectral features. Moreover, the losses of each combination based on multiscale interfusion are calculated via weighted average, which enables SSPN to avoid the excessive influence of single combination in the updating of model parameters. Four HSI public datasets and several comparison models are employed to validate the classification effect of SSPN. Experimental results show that SSPN achieves the highest overall accuracy (OA) in all datasets compared with other classification models, with 100%, 98.8%, 99.8%, and 98.7% on the datasets of Chikusei, Pavia University, Botswana, and Houston 2013, respectively. SSPN is demonstrated to possess higher classification accuracy and better generalization performance on HSI.**

***Index Terms*— 3-D convolutional neural network (3D CNN), feature pyramid structure, hyperspectral image (HSI) classification, multiscale convolutional extraction, multiscale interfusion, spatial–spectral pyramid network (SSPN).**

I. Introduction

HYPERSPECTRAL image (HSI) is characterized by high spectral resolution and a large number of spectral bands, which allows it to contain more information than ordinary image [1]. In recent years, with the rapid development of HSI technology, it has been applied in many fields, such as remote sensing images [2], [3], biomedicine [4], and agricultural production [5]. Most of the research content concentrates on the utilization of HSI classification to identify research object.

Predicting the category of each pixel point in the image is considered as the main task in the HSI classification. The traditional methods of HSI classification mainly include support vector machine (SVM) [6], k-nearest neighbor (K-NN) [7], random forest [8], and so on. However, high-dimensional characteristics and fewer training samples of HSI result in the difficulty of traditional classifiers to achieve high accuracy. For this reason, scientific researchers have proposed some methods for dimension reduction and spectral extraction, such as principal component analysis (PCA) [9], independent component analysis (ICA) [10], and linear discriminant analysis (LDA) [11]. HSI is typically nonlinear data; therefore, it is not properly processed using the above methods [12]. Although this problem is relieved by introducing kernel function-based methods to a certain extent [13], deep learning methods characterized by nonlinearity and adaptive weights can better meet the requirements of HSI classification.

Deep learning methods have developed rapidly in the last decade and have shown great potential in the field of image processing [14]. Chen et al. [15] first applied deep learning methods in the HSI classification by using stacked autoencoders (SAEs) to extract spatial and spectral features simultaneously for the effective enhancement of classification accuracy. In [16], the same group also used deep belief networks (DBNs) for feature extraction and classification of HSI. Mou et al. [17] analyzed hyperspectral pixels serving as sequential data and then used recurrent neural networks (RNNs) to classify the pixels. Mei et al. [18] designed a bidirectional long short-term memory (Bi-LSTM) network and combined it with attention mechanism for HSI classification. Among many deep learning methods, convolutional neural network (CNN) has received the most attention from researchers. Unlike other deep learning methods, CNN can effectively reduce the quantity of model parameters by extracting various features of input data based on local connectivity, and thus, it is widely used in the field of computer vision [19]. In addition, CNNs and their derivative networks are employed in the HSI classification. Alipourfard et al. [20] used subspace-based feature extraction to achieve the dimension reduction of HSI and further completed HSI classification by using CNN. Zhao and Du [21] proposed the balanced local discriminant embedding algorithm (BLDE), increasing the distinguishability between different categories by introducing

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a balanced objective, projecting spectral information into a low-dimensional representation, and finally using CNN for HSI classification. Yue et al. [22] extracted depth features by deep CNN and then combined with logistic regression classifier for final classification. Guo and Zhu [23] proposed a CNN-based spatial feature fusion model that intelligently integrates spatial information into spectral information for obtaining good classification results. Mei et al. [24] proposed a CNN-based transformer named group-aware hierarchical transformer (GAHT), and it used a new module grouped pixel embedding (GPE) to emphasize local relationships within HSI spectral channels. The convolutional kernel in CNN has various shapes. Tran et al. [25] proposed a 3D CNN model in 2015 for the first time. The convolutional kernel in 3D CNN model can move in space, namely, sliding on the channel dimension, and this characteristic is considered to be in accordance with HSI. For this reason, Chen et al. [26] first applied 3D CNN in the HSI classification. He et al. [27] proposed a multiscale 3-D deep CNN for HSI classification. Zhong et al. [28] designed a residual module based on 3D CNN to extract spatial and spectral information separately and also applied it in HSI classification. It is noteworthy that 3D CNN itself can extract spatial and spectral information at the same time.

Each of the HSI classification methods described above has its respective characteristics, but from the viewpoint of HSI characteristics, there are still room for further optimization and improvement.

1) Most models extract HSI spectral features in a single way, and thus, it is difficult to extract sufficient spectral features effectively. Can we design a structure that both enhances the extraction of spatial and spectral features and further enables the model to learn a sufficient quantity of spatial–spectral features?
2) In the processing of HSI spectral features, most models directly calculate the output class via fully connected layer. Although this approach can simplify the output process of the model and save significant time, it does not allow the model to learn the spectral features adequately and eventually causes the bottleneck of model accuracy. Therefore, can we design a framework that enables the model to learn enough spectral features for improving model accuracy?
3) 3D CNN is the most suitable deep learning model for HSI, but it does not have a deep and specific model framework. Can we combine the above two ideas to design a more complex 3D CNN structure that contributes to HSI classification?

To solve these problems, we designed a new 3D CNN-based classification model called the spatial–spectral pyramid network (SSPN). This model has several advantages as follows.

1) To solve the problem of spectral feature extraction, we gain inspiration from feature pyramid networks (FPNs) and further design a structure for multiscale extraction based on FPN. We take advantage of 3-D convolution kernel at three different scales to extract HSI spatial–spectral features simultaneously. Each type of convolution can extract spatial–spectral features differing from other scales and greatly enrich the diversity of the model for spatial–spectral feature learning.
2) In order to improve the learning ability of the model for spectral features, we propose a structure that contains two-by-two fusion between different scales in a top-down process. This structure contributes to the combination of feature maps at different scales, enriching the features contained in a single feature map and improving the model sensitivity for HSI spatial–spectral information. It also enables the model to better learn the features between HSI spectra, which further improves the accuracy of the model.
3) In backpropagation, we use weighted average to calculate the losses of each combination based on multiscale interfusion. This step can avoid the excessive influence of single combination in the updating of model parameters.

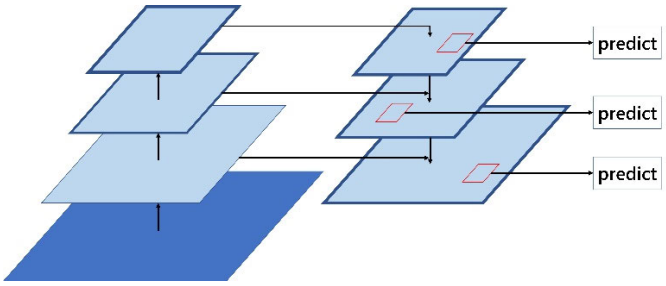


Fig. 1. Basic structure of the FPN.

The rest of this article is structured as follows. Section II outlines model-related works. Section III introduces the SSPN framework designed in this article. Section IV provides the experiment information in the public dataset and equipment. Section V presents the results and discussion. Section VI summarizes the application of SSPN in HSI classification.

II. Related Work

A. Feature Pyramid Network (FPN)

Lin et al. [29] proposed the FPN structure in 2017, which fused multistage feature maps for extracting both high-level semantic features and bottom-level contour features. Fig. 1 shows the basic structure of FPN.

The FPN structure is divided into two sections, including bottom-up feature extraction and top-down feature fusion. Bottom-up feature extraction is regarded as the forward propagation process of neural networks where feature maps are downsampled and different degrees of semantic information are obtained at each upper level. On the other hand, the top-down process is divided into two steps, including upsampling on feature maps at higher levels and fusing feature maps obtained by upsampling with feature maps of previous layer by lateral connection in order to enhance the features at high levels. The main advantage of FPN is located at the combination of feature information at different scales for enhancing the detection of different targets.

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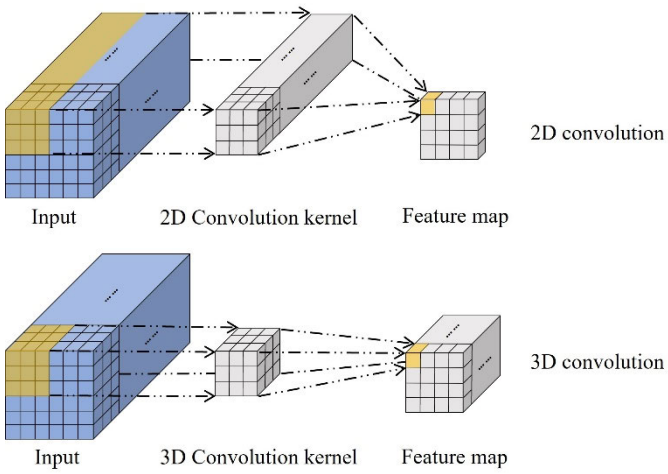


Fig. 2. Difference between 2-D convolution and 3-D convolution.

B. 3-D Convolution Kernel

Unlike 2-D convolution, the convolution kernel in 3-D convolution can slide on the channel. The 3D CNN model is capable of learning abundant features on the input data with multiple channels, which is perfectly suitable for HSI characteristics. Thus, the 3D CNN model can learn enriched spectral features of HSI for classification. Fig. 2 shows the difference between 2-D convolution and 3-D convolution.

In 3D CNN, the output value of neuron (x, y, z) is O_{ij}^{xyz} is calculated as follows:

$$O_{ij}^{zyz} = \left(\sum_{m} \sum_{p=0}^{K_w-1} \sum_{p=0}^{K_h-1} \sum_{p=0}^{K_c-1} w_{ijm}^{pqr} v_{(i-1)m}^{(x+p)(y+q)(z+r)} + b_{ij} \right) \quad (1)$$

where i and j represent the ith layer and the jth feature map, respectively, and K_w, K_h, and K_c represent the size of convolution kernel. m represents the number of convolution kernels. x, y, and z represent the location of feature map. w_{ijm}^{pq} and b_{ij} represent the weight at the position and the bias value of the jth feature map at the ith layer, respectively. f denotes the activation function. Commonly used activation functions include rectified linear unit (ReLU), tanh, sigmoid functions, and so on.

C. Spectral Data

An ordinary color image is regarded as a 3-D matrix with the size of $H \times W \times 3$, in which 3 represents the picture channel (red-green-blue (RGB) tricolor channels). Here, the concept of "spatial dimension" is defined as the 2-D representation of an image. More importantly, HSI has a greater number of spectral channels and unprocessed hyperspectral data when compared with normal images. In HSI, not only the spatial information provided by normal images is achieved but also abundant spectral data of each pixel point in the channel.

III. Spatial–Spectral Pyramid Network (SSPN)

A. Overall Architecture

FPN greatly enriches the feature contained in feature maps via two processes, including bottom-up and top-down pathways, allowing the model to learn different objects. The normal image provides the most important information from the spatial dimension, such as target size, spatial location, and edge contours. However, only using the spatial information is not enough for HSI classification and its spectral dimension is in analogy with the spatial dimension of normal image due to strong 3-D properties of HSI image. The 3-D convolution characterized by sliding on spectral dimension is selected as the extractor of spatial–spectral features. Based on these points mentioned above, SSPN is designed and applied in the HSI classification. It can be regarded as a double pyramid model that allows the extraction of pyramid structure both in spatial dimension and spectral dimension.

As shown in Fig. 3, the model sets up three different scales of 3-D convolution kernels to extract spatial–spectral features in parallel. As a result, SSPN learns the spectral features at different distances separately and uses the operation of two-by-two fusion to further enhance the spatial–spectral features for improving its learning ability for spectral information.

B. Different Subunits

1) Bottom-Up Pathway: The bottom-up process is a feed-forward computation using CNNs, in which each group uses the same 3-D convolutional kernel size and their feature maps are scaled by using maximum pooling. Three different sizes of convolution kernel are set to $3 \times 3 \times 3$, $3 \times 3 \times 5$, and $3 \times 3 \times 7$, as well as $h \times w \times c$ represent the sampling distance of convolution kernel in length, width, and channel, respectively. Thus, SSPN is able to extract the correlation between the spectrums at different distances based on different sizes of convolution kernel. Meanwhile, 3-D convolution is suitably chosen as the feature extractor because its characteristics with movement in 3-D matrix enables the model to diversify sampling patterns in the spectral dimension and its ability of extracting spectral and spatial features simultaneously also allows the model to learn spatial–spectral information. Fig. 4 shows the downsampling by bottom-up pathway at a kernel size of $3 \times 3 \times 3$. The same downsampling pattern is used in three different groups of feature extraction modules (labeled as blue, yellow, and green in Fig. 3), and the operation of convolutional kernel and max pooling uses $3 \times 3 \times c$ and $2 \times 2 \times 2$, respectively.

2) Top-Down Pathway: The top-down pathway is divided into two parts including upsampling and lateral connection, as shown in Fig. 5. The same pattern of top-down pathway is used in the feature fusion at three different scales.

a) Upsampling: In the process of bottom-up feature extraction, the top-level feature maps (A_5, B_5, C_5) includes condensed and different spatial–spectral information under three different settings of sampling size. In order to fuse top-level feature with other feature maps with different sizes for further enriching features contained in the feature map, we set a common channel quantity and used $1 \times 1 \times 1$ convolution kernels to adjust the quantity. They are employed to enable top-level feature maps and those obtained by sampling at other scales at equal channel quantity, and a trilinear interpolation method is then utilized to upsample on high-level features for allowing high-level feature map $(A_5, B_5, C_5, D_1D_3, E_1E_3, F_1F_3)$ and those obtained by sampling at other scales of equal size (except for non-top-level feature maps). Since the results based on 3-D convolution

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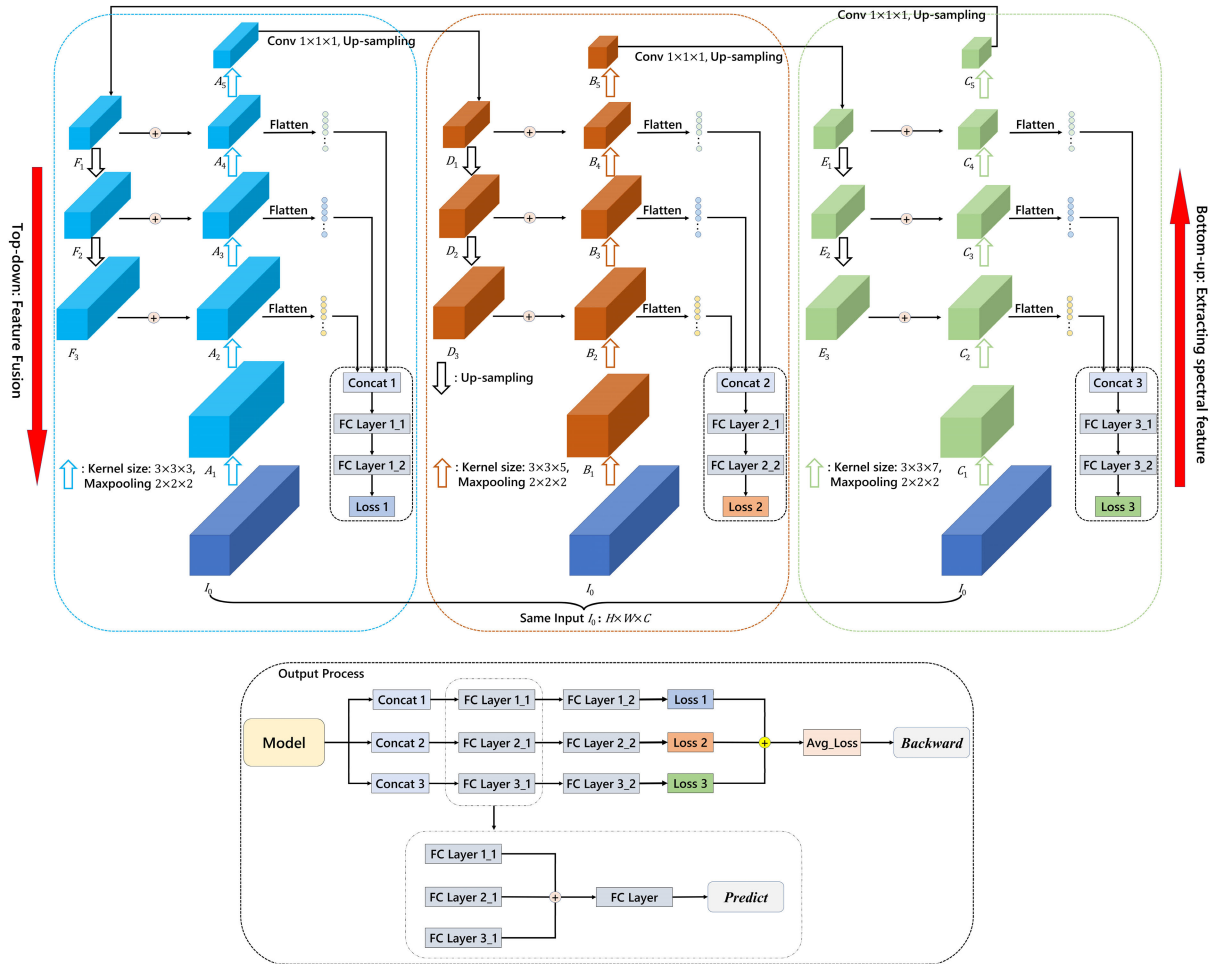


Fig. 3. Structure of SSPN.

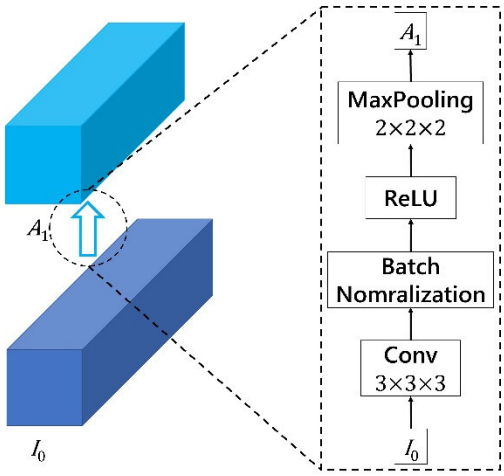


Fig. 4. Bottom-up pathway of SSPN.

are upsampled, the model proposed in this article ingeniously constitutes a double pyramidal structure at spatial–spectral dimensions.

b) Lateral connection: Fig. 5 shows the process of lateral connection. After upsampling, feature maps at different scales are fused by element addition to complete the operation of lateral connection. Its purpose aims at the fusion of features obtained by using convolutional kernels at different scales to sample and further enhancement of features contained in the feature map.

Overall, both high-level features and bottom-up features are enhanced via upsampling and lateral connection. Unlike the general way of introducing the FPN structure into the HSI classification model, SSPN makes use of 3-D convolution to achieve the FPN structure and then performs the feature extraction on HSI by using 3-D convolution kernel at multiple scales that reflect different scales in spectral dimension. This design taking advantage of 3-D convolution characteristics contributes to the combination between spatial information extracted by the model and spectral information at different distances, thus improving the learning ability of the model for HSI features. Moreover, lateral connection using two-by-two fusion at different scales intends to achieve the interactive influence of spatial–spectral features at different scales and then increase the model sensitivity to spatial–spectral information, resulting in better improvement of HSI classification effect.

3) Fully Connected Layer and Output: The structure of fully connected layers is designed to facilitate the model to calculate the subsequent classification results, as shown in Fig. 6.

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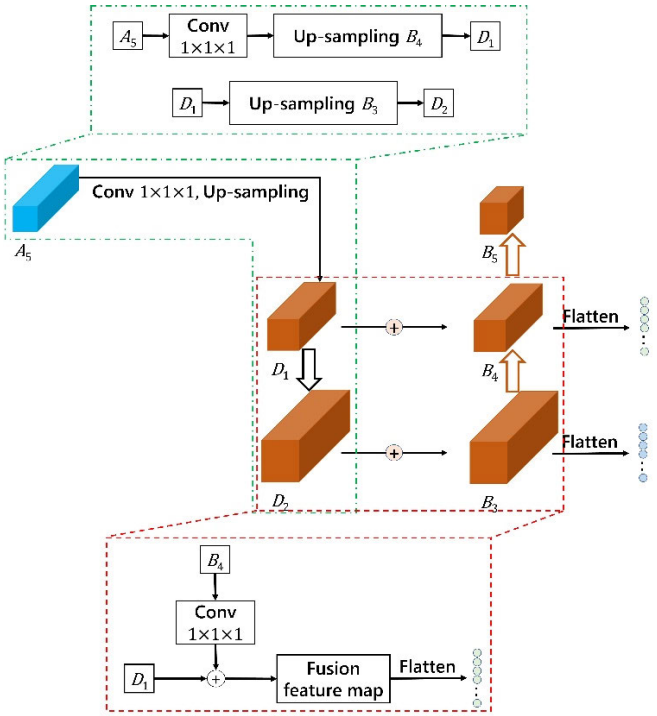


Fig. 5. Top-down pathway of SSPN.

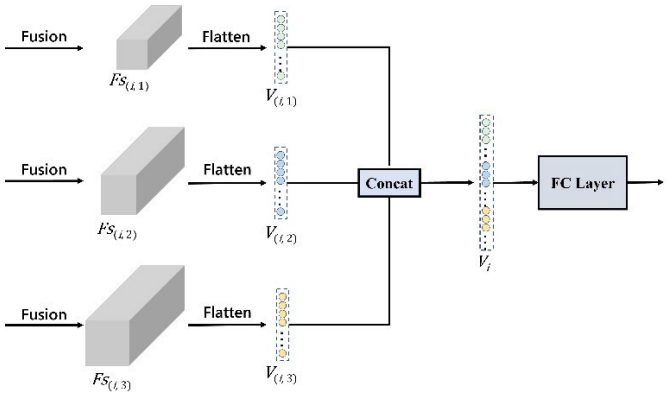


Fig. 6. Structure of each group of fully connected layers.

After the flatten process of fused feature maps $Fs_{(i,j)}$, each set of fused features yields three of 1-D vectors $V_{(i,j)} = \{V_{i,1}, V_{i,2}, V_{i,3}\}$, $i = 1, 2, 3$, where i represents the group and j represents the vectors within the group. A 1-D vector V_i is calculated by the Concat operation and then combined with two fully connected layers to output the results of each group. Finally, the loss of three sets of results is calculated by cross entropy and represents $L_i = L_1, L_2, L_3$. As the results of three groups are calculated by interactive fusion, the average value of their losses is used to obtain the final loss L_{avg} that further participates in the backpropagation.

IV. Experiment Preparation

A. Experimental Dataset

Four public datasets, including Pavia University dataset, Botswana dataset, Chikusei dataset, and Houston 2013 dataset, were used in the experiment. The training set of each dataset is 30% of the total number of samples.

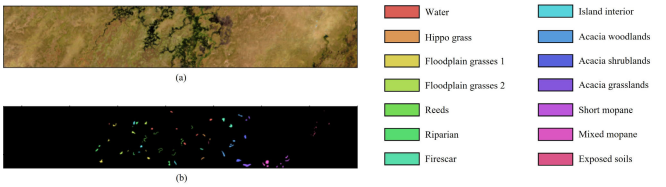


Fig. 7. Pseudo-color images (R: 75, G:33, and B: 15) and ground truth of Botswana dataset.

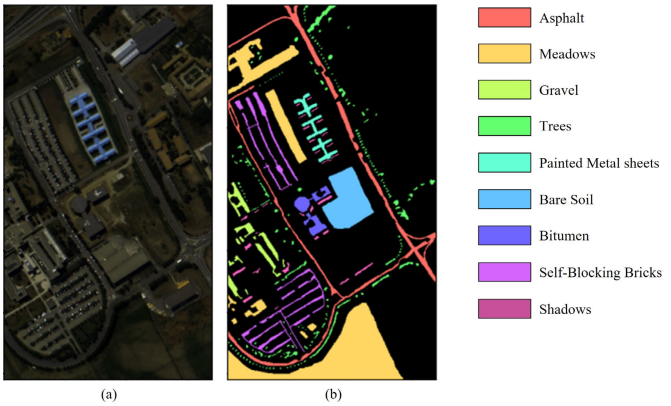


Fig. 8. Pseudo-color images (R: 60, G:30, and B: 2) and ground truth of Pavia University dataset.

1) Botswana Dataset: Fig. 7 shows the pseudo-color images and ground truth in the Botswana dataset. This dataset presents a series of data from the NASA with the EO-1 satellite at Botswana. In 2001, the EO-1 satellite acquired 14 categories of ground truth such as water and reeds, with 30-m pixel resolution in 242 bands covering 400–2500 nm. After removing noise bands, the remaining 145 bands became candidate features. The Botswana dataset is available at https://www.ehu.eus/ccwintco/index.php?title=Hyperspectral_Remote_Sensing_Scenes.

2) Pavia University Dataset: Fig. 8 shows the pseudo-color images and ground truth in the Pavia University dataset. This dataset was acquired with the ROSIS sensor at Pavia city, Italy, in 2002, and includes nine categories of ground truth such as roads, trees, and roofs. The image size is 610 × 340 pixels with a spatial resolution of 1.3 m. There are 115 bands in the wavelength range of 430–860 nm, of which 103 bands are reserved for testing after removing 12 bands with strong noise and water vapor absorption. The Pavia University dataset is available at https://www.ehu.eus/ccwintco/index.php?title=Hyperspectral_Remote_Sensing_Scenes.

3) Chikusei Dataset: Fig. 9 shows the pseudo-color images and ground truth in the Chikusei dataset. This dataset, including 19 categories of ground truth such as water and forest, was obtained by Headwall Hyperspec-VNIR-C in Chikusei, Japan, in 2014 [30]. The resolution size of the image is 2517 × 2335, and the spatial resolution is 2.5 m. The dataset contains 128 bands in the range of 343–1018 nm.

4) Houston 2013 Dataset: Fig. 10 shows the pseudo-color images and ground truth in the Houston 2013 dataset. This dataset was provided by the 2013 IEEE GRSS Data Fusion Competition with the ITRES CASI-1500 sensor. The data, including 15 categories of ground truth such as water and road,

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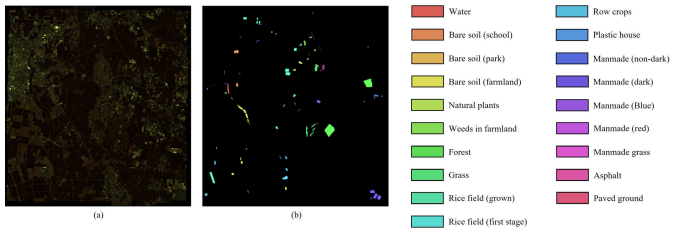


Fig. 9. Pseudo-color images (R: 43, G:21, and B: 11) and ground truth of Chikusei dataset.

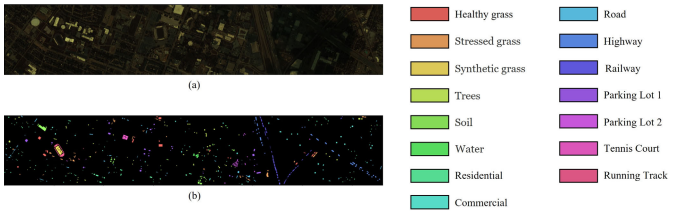


Fig. 10. Pseudo-color images (R: 43, G:21, and B: 11) and ground truth of Houston 2013 dataset.

TABLE I
EFFECT OF DIFFERENT PATCH SIZES ON SSPN ACCURACY (%)

Patch Size	Pavia University	Botswana	Chikusei	Houston 2013
9×9	**98.808**	**99.795**	**99.994**	**98.692**
11×11	98.699	99.179	99.695	98.448
13×13	98.609	99.333	99.695	97.505
15×15	97.970	97.999	99.555	96.829
17×17	97.062	97.075	99.272	94.777

possess the pixel size of 349 × 1905 and contains 144 bands in the range of 364–1046 nm. The Houston 2013 dataset is available at http://www.grss-iee e.org/community/technical-committees/data-fusion/.

B. Experimental Equipment

In the equipment configuration, the CPU is Intel Xeon Gold 6230R with 2.10 GHz × 104, the RAM is 256 GB, the graphics card is Nvidia A30 with 24 GB of video memory, the operation system is Ubuntu 18.04, and the deep learning tool is Pytorch.

V. EXPERIMENTAL RESULTS AND DISCUSSION

A. Effect of Patch Size on SSPN

Table I shows the effect of different patch sizes on the accuracy of SSPN. SSPN has the highest overall accuracy (OA) on all datasets when the size of patch size is 9 × 9, and the accuracy decreases gradually with the augment of patch size. However, the minimum of patch size is determined to be 9 × 9 due to the setting of the SSPN model.

B. Ablation Experiment

An ablation experiment is performed to show that the effect obtained at the multiscale structure is the result of the combined effect at all three scales. We set up six comparison test models: single-scale 3-D feature pyramid structure (convolution kernel sizes with 3 × 3 × 3, 3 × 3 × 5, and 3 × 3 × 7) and double-scale 3-D feature pyramid structure (combination of convolution kernel sizes with 3 × 3 × 3 and 3 × 3 × 5, 3 × 3 × 3 and 3 × 3 × 7, and 3 × 3 × 5 and 3 × 3 × 7). The Houston 2013 dataset was selected as the dataset for this experiment because it has a more complex distribution of objects and a larger number of categories. The OA and AA are shown in Table II and Fig. 11.

TABLE II
RESULTS OF ABLATION EXPERIMENT ON HOUSTON 2013 (%)

Combination of kernel sizes	Model	OA	AA
Single	3	97.7	97.7
	5	96.6	96.9
	7	97.9	97.9
Double	3&5	97.8	97.5
	3&7	98.2	98.1
	5&7	98.4	98.4
Proposed	SSPN	98.7	98.6

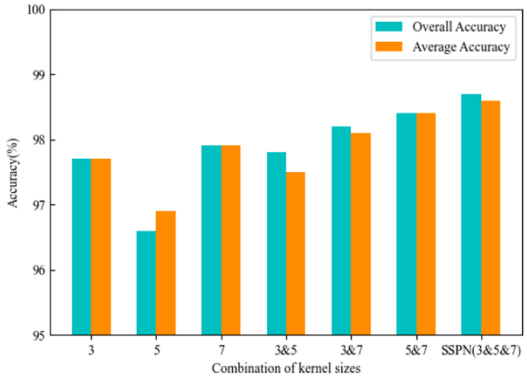


Fig. 11. Results of ablation experiment.

Table II and Fig. 11 show the results of ablation experiment. We can see a trend in Fig. 11. The OA of combinations 3 and 5 is higher than that of single kernel size 3 and 5, and the OA of combinations 3 and 7 is higher than that of single kernel size 3 and 7. Therefore, we can get an explanation about this situation, and when different scales of convolution kernel combine, the OA and AA can increase obviously. Also, we can find that the convolution kernel 3 × 3 × 7 plays the most important role. In summary, the idea of our model is innovative.

C. Method Comparison

1) Introduction of Comparison Model: This article selects seven different models in the comparative experiment in order to demonstrate the innovation and effectiveness of SSPN. These models, including traditional models and the latest deep learning models in the field of HSI classification, are shown as follows.

1) *SVM:* The SVM model uses a radial basis kernel function (RBF kernel) and performs the HSI classification via raw spectral features.
2) *1-D CNN:* The 1-D CNN with structure as mentioned in [31] is implemented by using Pytorch and trained by the stochastic gradient descent (SGD) algorithm at a learning rate of 0.01.

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TABLE III
ACCURACY OF DIFFERENT CLASSIFICATION MODELS ON THE CHIKUSEI DATASET (%)

Class	SVM	1D CNN	RNN	SF	3D CNN	3D+1D Net	HybridSN	HF	**SSPN**
Water	99.2	99.8	**100**	99.9	98.9	**100**	99.2	**100**	**100**
Bare Soil (school)	97.6	99.5	99.9	99.9	96.7	**100**	99.9	**100**	**100**
Bare Soil (park)	1.0	41.9	95.2	68.8	93.5	98.9	88.5	97.3	**99.4**
Bare Soil (farmland)	94.7	97.7	99.7	97.8	94.8	99.9	99.1	**100**	**100**
Nature Plants	98.8	99.4	**100**	99.6	**100**	**100**	99.6	**100**	**100**
Weeds in farmland	90.4	96.0	99.5	98.8	96.1	99.8	95.3	95.1	**100**
Forest	99.4	99.5	99.9	99.9	99.9	**100**	**100**	**100**	**100**
Grass	99.3	99.3	**100**	99.7	99.6	99.9	99.4	99.9	**100**
Rice Field (grown)	99.7	99.5	99.9	99.8	98.5	**100**	98.3	99.0	**100**
Rice Field (first stage)	99.8	99.8	**100**	99.9	**100**	**100**	**100**	**100**	**100**
Row Crops	98.9	99.2	99.9	99.9	98.8	99.9	**100**	**100**	**100**
Plastic House	96.1	98.7	99.0	98.5	87.7	99.9	97.5	99.9	**100**
Manmade (non-dark)	97.5	98.9	99.3	98.8	92.8	99.9	99.1	**100**	**100**
Manmade (dark)	98.9	99.6	99.8	99.8	99.4	**100**	**100**	**100**	**100**
Manmade (blue)	**100**	**100**	**100**	98.0	97.1	**100**	96.6	**100**	**100**
Manmade (red)	**100**	**100**	**100**	**100**	83.5	**100**	82.2	95.7	**100**
Manmade Grass	97.7	99.3	**100**	99.9	99.5	99.9	**100**	**100**	99.9
Asphalt	87.6	95.8	97.0	97.6	56.0	99.5	64.7	87.5	**100**
Paved Ground	1.0	88.1	**100**	97.6	91.6	98.3	98.9	**100**	99.4
AA	87.2	95.4	99.4	97.6	93.9	99.7	95.7	98.6	**99.9**
OA	98.2	99.1	99.8	99.4	98.1	99.9	98.9	99.6	**100**
Kappa×100	0.980	0.989	0.998	0.994	0.978	0.999	0.988	0.995	**1.00**

TABLE IV
ACCURACY OF DIFFERENT CLASSIFICATION MODELS ON THE PAVIA UNIVERSITY DATASET (%)

Class	SVM	1D CNN	RNN	SF	3D CNN	3D+1D Net	HybridSN	HF	**SSPN**
Asphalt	84.9	93.7	94.1	95.9	68.9	99.0	77.9	98.8	**99.6**
Meadows	93.1	95.5	96.8	96.7	95.8	**99.2**	95.8	99.1	99.1
Gravel	53.7	80.2	73.2	86.0	87.0	97.9	90.0	99.2	**99.6**
Trees	92.8	91.9	96.4	95.4	56.3	**99.1**	58.4	94.5	97.9
Painted Metal Sheets	99.4	99.5	**99.9**	99.4	92.9	99.5	92.3	92.7	98.8
Bare Soil	70.6	89.1	92.7	92.8	97.4	99.5	98.0	**100**	99.9
Bitumen	1.2	87.3	86.1	91.3	96.7	97.5	95.6	99.3	**99.9**
Self-Blocking Bricks	82.5	86.9	76.3	89.4	90.7	97.9	89.5	98.5	**99.2**
Shadows	**100**	99.8	99.8	99.8	36.1	98.6	49.6	83.2	97.3
AA	75.4	91.6	90.6	94.1	80.2	98.7	83.0	96.1	**99.0**
OA	86.1	92.4	92.4	94.5	86.2	98.7	88.4	97.9	**98.8**
Kappa×100	0.810	0.899	0.900	0.928	0.820	0.982	0.846	0.973	**0.984**

3) *RNN:* The RNN model similar to that in [17] is used for HSI classification, and the activation function is replaced with the commonly used tanh function and Pytorch is used to implement the model.
4) *Spectral Former (SF):* The SF model with structure as shown in [32] is improved based on the transformer by adding two new modules, including group-wise spectral embedding (GSE) and cross-layer adaptive fusion (CAF), and implemented by using Pytorch.
5) *3D CNN:* The 3D CNN model uses a conventional structure proposed in [26], which is composed of three convolution-pooling layers and one fully connected layer. This model is implemented with Pytorch and trained by SGD at a learning rate of 0.003.
6) *3-D + 1-D Net:* The 3-D + 1-D Net model is implemented by using Pytorch according to the structure provided in [33] and trained by SGD at a learning rate of 0.01. This model named 3-D + 1-D Net is characterized by using the 1-D convolution instead of the commonly used pooling method and fully connected layer.
7) *HybridSN:* The model similar to the structure in [34] is implemented with Pytorch. The patch size is 25 × 25. HybridSN consists of two fully connected layers and four convolutional layers, including three 3-D convolutional layers for spatial–spectral features and one 2-D convolutional layer for spatial features.
8) *Hybrid Former (HF):* The HF model is implemented by using Pytorch, and the code is provided in [35]. The model proposed a hybrid transformer that integrates multigranularity tokens and spatial–spectral attention and a transformer encoder with an SSA mechanism. The patch size of HF is 15.

2) Results Discussion: Tables III–VI show the OA, AA, and kappa coefficients of eight different classification models

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TABLE V
ACCURACY OF DIFFERENT CLASSIFICATION MODELS ON THE BOTSWANA DATASET (%)

Class	SVM	1D CNN	RNN	SF	3D CNN	3D+1D Net	HybridSN	HF	**SSPN**
Water	**100**	**100**	99.7	**100**	98.8	**100**	82.7	96.6	99.7
Hippo Grass	93.9	97.6	96.8	97.5	98.3	**100**	96.1	**100**	**100**
Floodplain Grasses 1	86.9	89.9	99.0	96.3	**100**	**100**	91.1	99.7	**100**
Floodplain Grasses 2	81.0	87.0	94.4	95.0	97.0	95.2	78.7	**100**	98.9
Reeds	79.0	83.4	88.0	85.8	95.2	94.1	75.7	97.2	**98.7**
Riparian	62.3	67.8	85.3	77.0	92.7	99.4	71.0	**100**	**100**
Firescar	97.4	98.4	99.0	99.0	98.4	**100**	**100**	**100**	**100**
Island Interior	83.6	87.7	98.4	95.8	99.6	**100**	56.3	**100**	**100**
Acacia Woodlands	78.5	83.9	92.6	86.2	99.7	99.7	99.7	**100**	**100**
Acacia Shrublands	74.8	80.0	94.2	87.7	**100**	97.4	94.6	**100**	**100**
Acacia Grasslands	92.3	94.0	94.5	93.2	94.8	98.1	95.4	**100**	**100**
Short Mopane	93.7	91.2	92.7	93.7	84.7	98.6	97.2	**100**	**100**
Mixed Mopane	83.9	84.9	93.1	89.0	93.4	**100**	98.2	**100**	**100**
Exposed Soils	97.3	95.8	98.3	97.4	**100**	**100**	88.2	99.1	**100**
AA	86.0	88.7	94.7	92.4	96.6	98.7	87.5	99.5	**99.8**
OA	85.0	88.0	94.3	91.7	95.1	98.7	88.3	99.4	**99.8**
Kappa×100	0.838	0.870	0.938	0.910	0.947	0.986	0.873	0.994	**0.998**

TABLE VI
ACCURACY OF DIFFERENT CLASSIFICATION MODELS ON THE HOUSTON 2013 DATASET (%)

Class	SVM	1D CNN	RNN	SF	3D CNN	3D+1D Net	HybridSN	HF	**SSPN**
Healthy Grass	90.8	96.0	91.3	97.4	85.7	**99.9**	85.8	97.5	98.6
Stressed Grass	89.7	98.3	89.4	97.6	71.4	98.2	69.3	98.2	**99.1**
Synthetic Grass	99.0	**99.9**	99.8	99.8	77.2	98.3	86.3	99.4	99.8
Trees	95.8	97.2	95.4	99.1	64.7	**99.1**	61.4	96.6	98.7
Soil	94.2	96.6	91.7	98.1	91.8	**100**	97.7	**100**	99.8
Water	91.4	99.2	99.5	99.0	91.7	94.0	85.3	94.3	**100**
Residential	74.2	77.6	56.1	88.1	79.2	94.0	32.4	97.8	**98.5**
Commercial	72.4	81.3	72.0	89.9	84.8	98.6	81.7	**99.7**	98.2
Road	71.5	74.0	59.9	83.8	72.6	95.0	57.0	97.2	**97.5**
Highway	60.6	83.2	58.2	89.9	89.3	96.4	95.5	**99.1**	98.6
Railway	63.4	73.3	63.6	86.3	82.7	96.4	81.6	99.5	**99.7**
Parking Lot 1	61.2	75.5	44.0	82.9	93.6	98.6	89.3	**99.5**	98.4
Parking Lot 2	15.1	43.7	47.1	63.0	80.2	**96.8**	78.2	96.7	93.0
Tennis Court	87.5	97.3	98.1	99.2	95.9	99.8	86.0	**100**	**100**
Running Track	99.2	98.7	99.1	**99.9**	81.4	99.5	85.7	99.4	99.7
AA	77.7	86.1	77.7	91.6	82.8	97.6	78.2	98.3	**98.6**
OA	78.9	86.4	75.8	91.7	82.0	97.7	78.0	98.4	**98.7**
Kappa×100	0.771	0.853	0.738	0.910	0.805	0.975	0.762	0.984	**0.986**

on four public datasets. From the experimental results, the SSPN model proposed in this article achieves the optimal classification accuracy with 100%, 98.8%, 99.8%, and 98.7% on the Chikusei, Pavia University, Botswana, and Houston 2013 datasets, respectively. It is noteworthy that the Chikusei dataset with the maximum volume of data leads to approximately the same OA based on all models. Specifically, SVM has a low value of OA on four datasets with 53.7% accuracy on Gravel of Pavia University dataset and 62.3% accuracy on Riparian of Botswana dataset because of weak learning ability of SVM on features compared with other deep learning models. 1-D CNN, RNN, and SF that all utilize a single pixel point for training account for 86.4%, 75.8%, and 91.7% OA on the Houston 2013 dataset, respectively, while SSPN using spatial information represents a higher value of OA with 98.7% on the same dataset. As for other three models utilizing spatial information, 3D CNN shows a lower value of OA with 86.2% on the Pavia University dataset when compared with SSPN, which is explained by the influence of redundant spectral information of HSI in model training and without targeted learning on the features of HSI. Although 3-D + 1-D Net replaces traditional pooling and fully connected layers with 1-D convolution for improving its learning ability on HSI, 3-D + 1-D Net lacking of targeted learning on the features of HSI produces slightly lower OA with 98.7% on the Botswana dataset when compared with SSPN.

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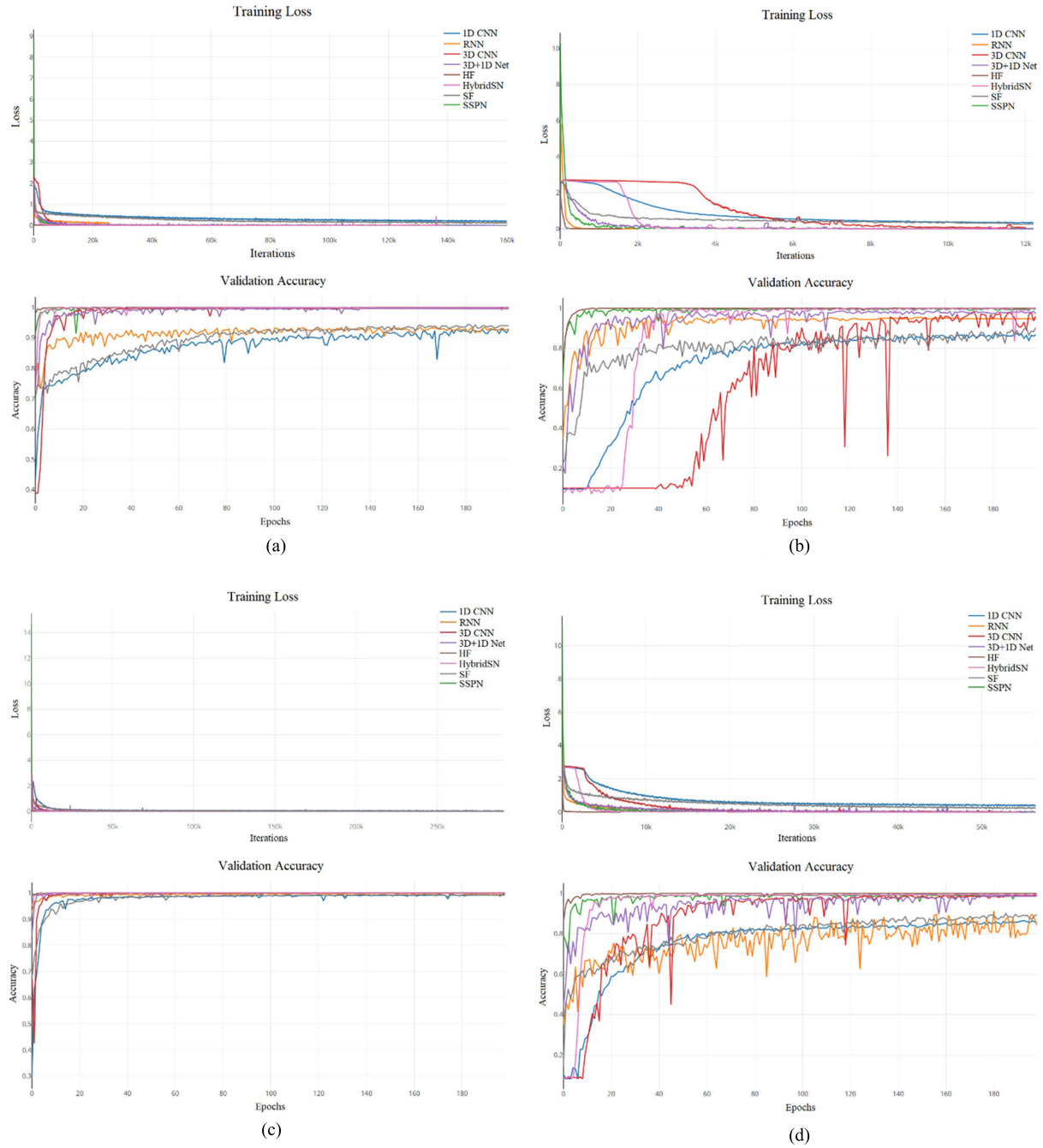


Fig. 12. Training loss and validation accuracy curves of SSPN and six deep learning models on four public datasets. (a) Pavia University. (b) Botswana. (c) Chikusei. (d) Houston 2013.

Moreover, HybridSN strengthening learning ability on spatial features provides a very lower value of OA with 78.0% on the Houston 2013 dataset due to its weak learning ability on spectral information. HF is the state-of-the-art model for HSI classification, so it has a good performance on classification. Its accuracies on three datasets are close to our model and 1% less accurate than us on the Pavia University dataset. However, although HF has a good performance, it also has a mass of hyperparameters. In contrast, our model requires fewer hyperparameters and has better robustness and generalizability. Table VII shows the difference between nine models. To sum up, SSPN proposed in this article uses the 3-D convolution to extract spatial information in combination with the exaction and fusion of spectral dimension at multiple scales, enabling it to learn abundant spatial–spectral features. SSPN also learns different classes of spectral features due to its multiscale learning on the same region in spectral dimension. Therefore, SSPN achieves the highest OA on all datasets for HSI classification compared with other models.

In order to describe the advantage of our model, we have counted the hyperparameters in all models, and it is shown in Table VII.

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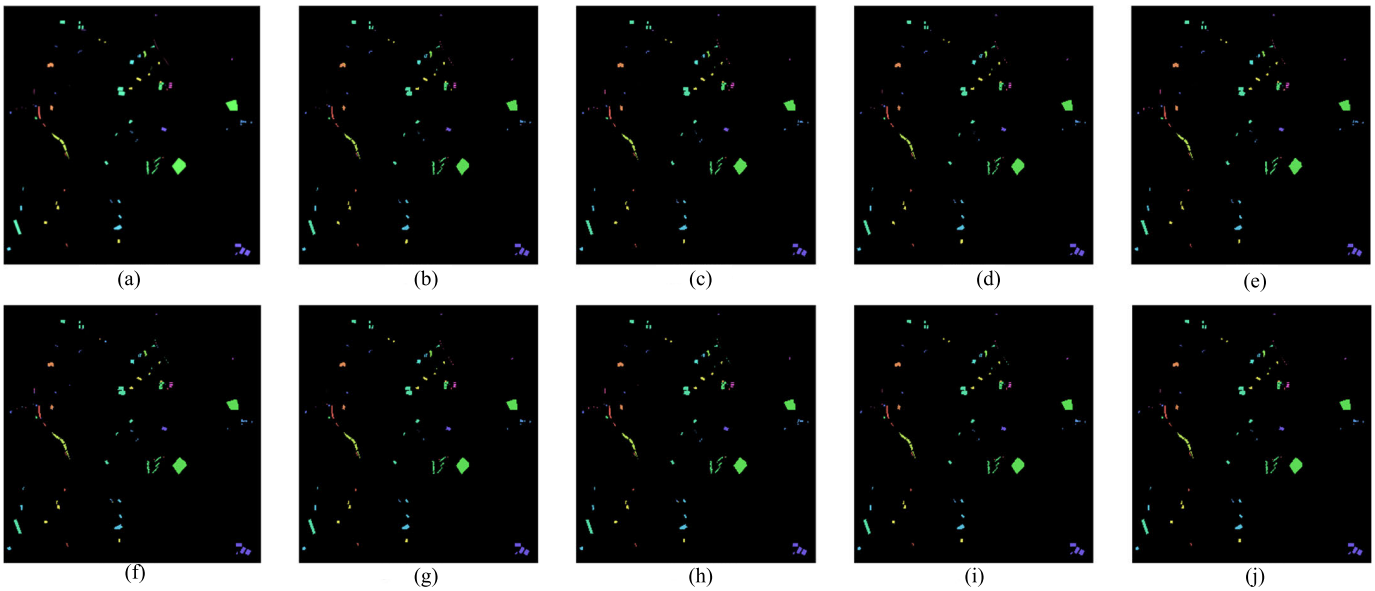


Fig. 13. Ground-truth map and model classification maps of Chikusei dataset. (a) Ground truth. (b) SVM. (c) 1-D CNN. (d) RNN. (e) SF. (f) 3D CNN. (g) 3-D + 1-D Net. (h) HybridSN. (i) HF. (j) SSPN.

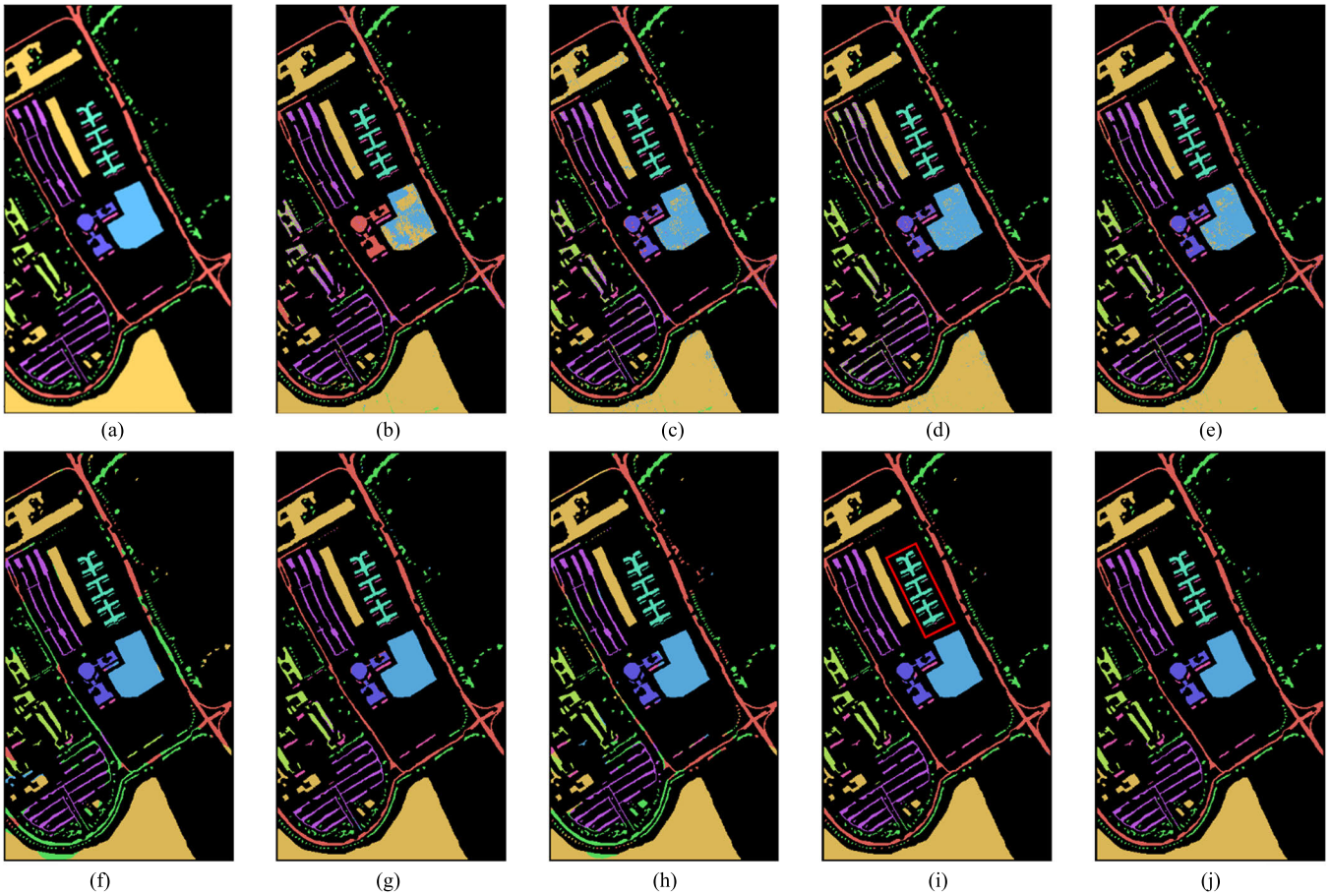


Fig. 14. Ground-truth map and model classification maps of Pavia University dataset. (a) Ground truth. (b) SVM. (c) 1-D CNN. (d) RNN. (e) SF. (f) 3D CNN. (g) 3-D + 1-D Net. (h) HybridSN. (i) HF. (j) SSPN.

From Table VII, we can find that the hyperparameters of our model are lower than other models. It can be proved that our model has good robustness and generalization.

Fig. 12 shows the training loss and validation accuracy of SSPN and other six deep learning models on four public datasets. The figure of training loss on dataset shows that the

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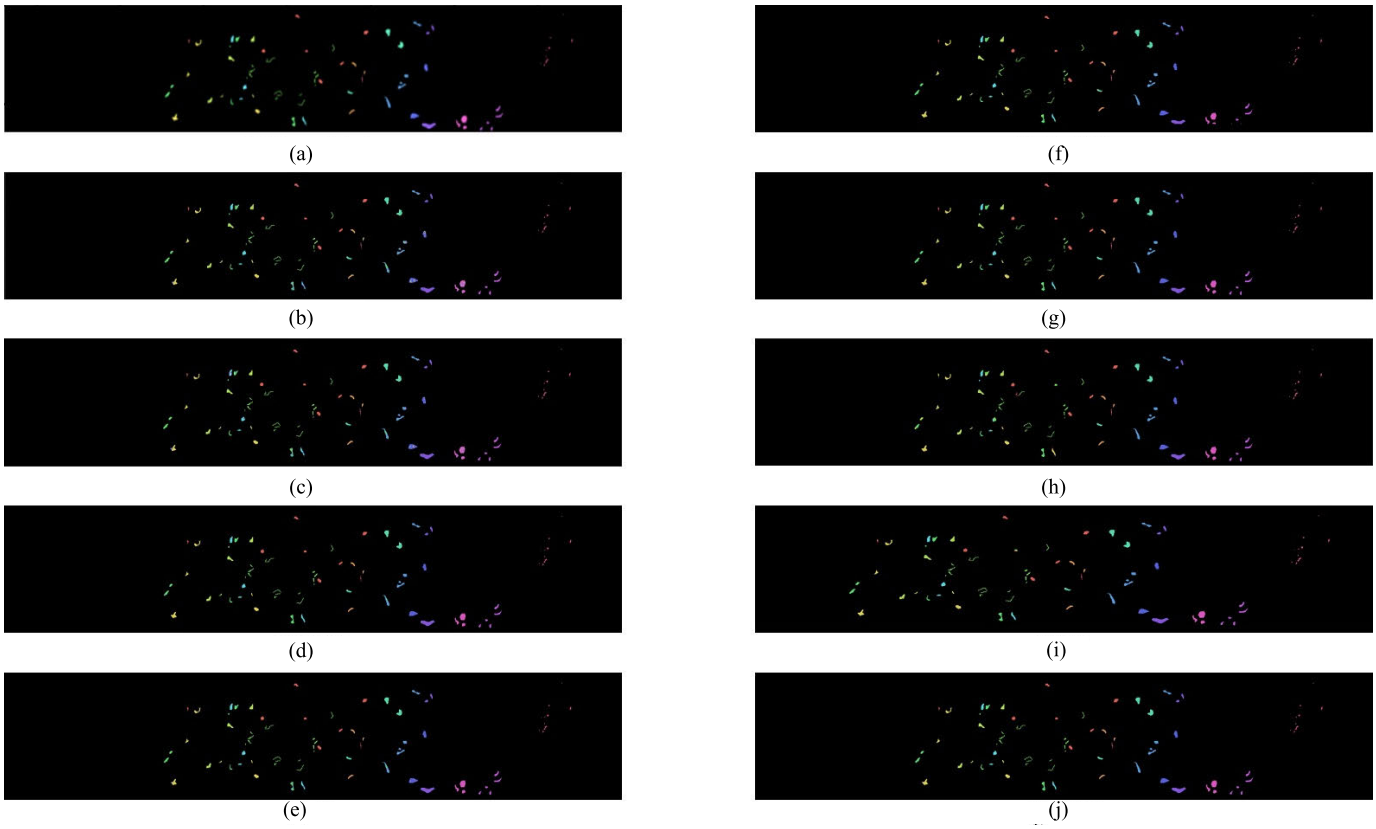


Fig. 15. Ground-truth map and model classification maps of Botswana dataset. (a) Ground truth. (b) SVM. (c) 1-D CNN. (d) RNN. (e) SF. (f) 3D CNN. (g) 3-D + 1-D Net. (h) HybridSN. (i) HF. (j) SSPN.

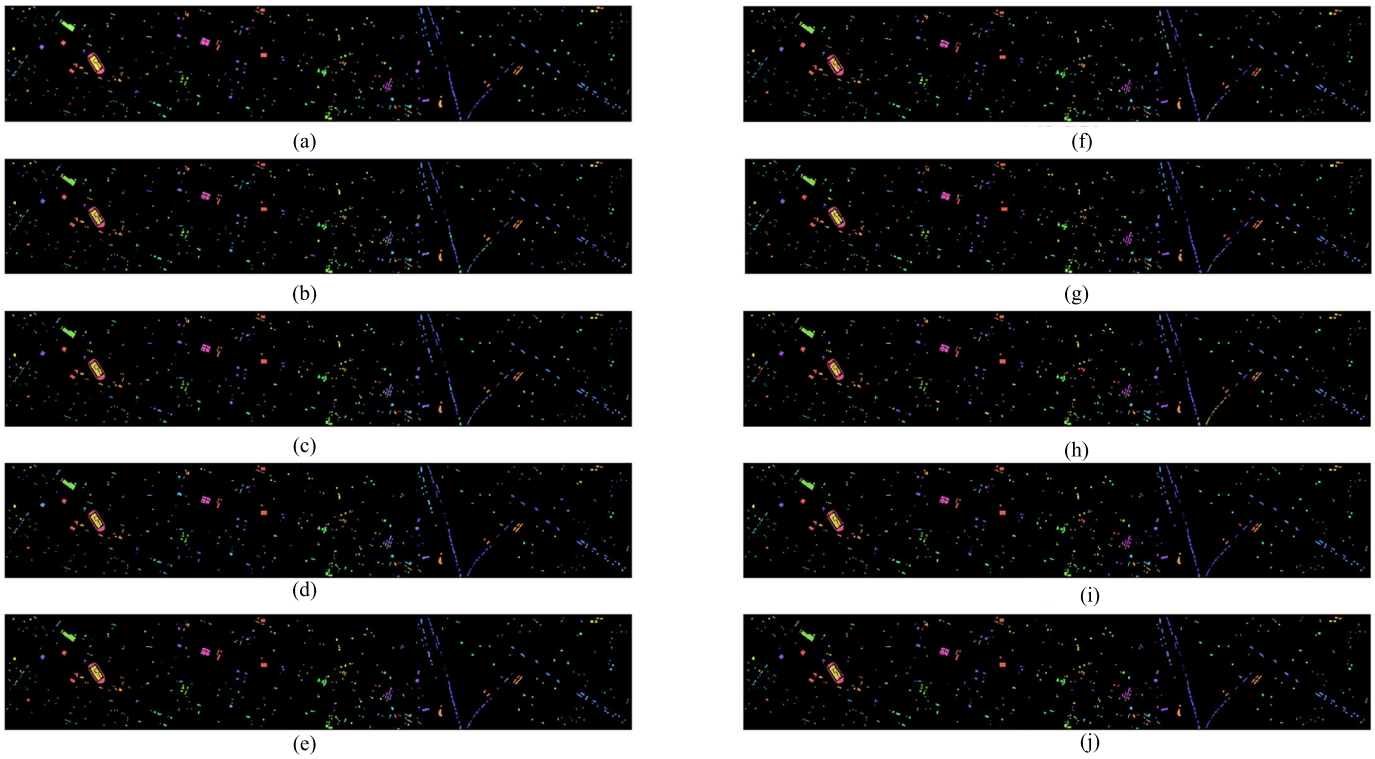


Fig. 16. Ground-truth map and model classification maps of Houston 2013 dataset. (a) Ground truth. (b) SVM. (c) 1-D CNN. (d) RNN. (e) SF. (f) 3D CNN. (g) 3-D + 1-D Net. (h) HybridSN. (i) HF. (j) SSPN.

convergence rates of SSPN and HF are significantly higher than that of other models.

In addition, SSPN and HF have the highest verification accuracy in the verification accuracy graph. HF is the state-of-the-art model, so it has advantages in feature extraction and processing. However, we combined with comparative analysis in Tables III–VI, and SSPN shows superior advantage in the HSI classification.

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TABLE VII
RESULTS OF ABLATION EXPERIMENT ON HOUSTON 2013 (%)

Model	Number of hyperparameters
SVM	5
1D CNN	4
RNN	4
SF	12
3D CNN	4
3D+1D Net	5
HybridSN	6
HF	12
SSPN	4

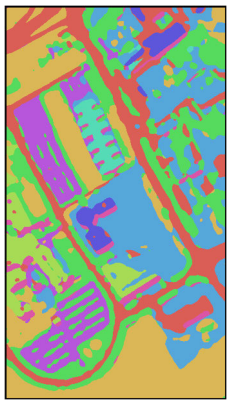


Fig. 17. Classification map with all pixels on Pavia University (OA: 98.8%).

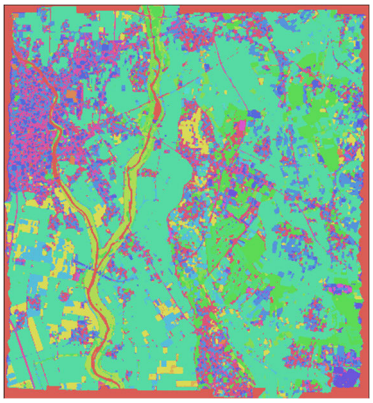


Fig. 18. Classification map with all pixels on Chikusei (OA: 100%).

Figs. 13–16 show the classification maps of SSPN and other models on four public datasets. SVM, 1-D CNN, RNN, and SF that all use the spectrum of single pixel points for learning produce obvious misclassification points, while SSPN has fewer regions of misclassification compared with other three models, including 3D CNN, 3-D + 1-D Net, and HybridSN with the setting of patch size because it uses multiscale extraction and fusion of spectral dimension for improving the model sensitivity to spatial–spectral information when extracting spatial–spectral features. The classification map of HF has obvious misclassification points on the Pavia University dataset, and we labeled them with a red box. SSPN learns spatial–spectral information in model training and further obtains better classification results in HSI. Thus, SSPN proposed in this article is preferably selected for HSI classification in comparison with other models.

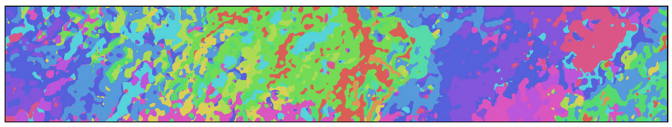


Fig. 19. Classification map with all pixels on Botswana (OA: 99.8%).

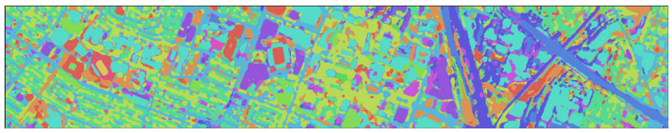


Fig. 20. Classification map with all pixels on Houston 2013 (OA: 98.7%).

In order to illustrate the effect of our model on edge preserving at all pixel points, we performed the experiment on all datasets. The results are shown in Figs. 17–20.

In Figs. 17–20, we can see that our model has the good effect on edge preserving. SSPN can effectively classify the objects presented in the pseudo-color images. It further illustrates that our model has the good classification performance.

VI. CONCLUSION AND FUTURE WORK

In this article, a novel 3D CNN model, SSPN, combining the characteristics of FPN, 3-D convolution, and feature fusion is applied in the HSI classification. More specifically, SSPN using 3-D convolution innovatively designs the spectral dimension at multiple scales, allowing it to extract abundant spatial–spectral features in bottom-up pathway. SSPN also integrates with two-by-two fusion for increasing the model sensitivity to spatial–spectral information of HSI, which enables the model to better learn the spatial–spectral features and further enhances the learning ability. In addition, SSPN calculates the loss average of each group with the same kernel size, enabling it to avoid the excessive influence of single combination in the updating of model parameters. SSPN can significantly improve the performance of HSI classification, thus becoming an emerging model in the mainstream of classification algorithm. We demonstrate the effect of SSPN by comparing with other models on four public datasets. In future research, we hope that the interpretability of SSPN is able to be further improved according to the priori theory of HSI mechanism, and SSPN with innovative structures inspires the establishment and development of complex 3D CNN models.

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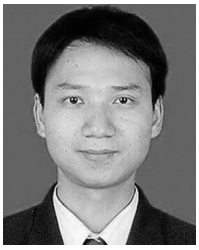
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